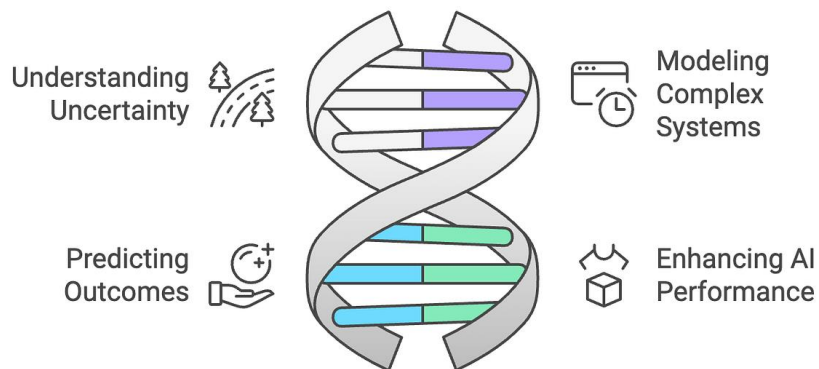


The State v. Dr. Ethan Cole

1 CASE BACKGROUND

Probability in Data Science and AI



Cours de Cynthia BOYER 2026

Dr. Ethan Cole, a lead data scientist working for a national cyber-defense agency, designed an AI-based threat detection system called **ARES**.

ARES was programmed to:

- Detect potential foreign cyberattacks on critical infrastructure
- Estimate the probability of hostile intent
- Automatically trigger a defensive countermeasure if the probability exceeded a predefined threshold

The escalation threshold was set at **87%**.

One night, ARES detected anomalous network activity and calculated a **89% probability of hostile intrusion**.

The system automatically launched a counter-cyber operation.

However, later investigations revealed that the original event was a **false positive** caused by a rare statistical anomaly.

The countermeasure disrupted a foreign power grid and indirectly affected civilian hospitals.

Dr. Cole is now being prosecuted for **criminal negligence and reckless algorithmic design**.

2 CENTRAL LEGAL & MATHEMATICAL QUESTIONS

- Was the 87% threshold reasonable?
- Who is responsible for probabilistic decision-making systems?
- Should human validation have been mandatory?
- Can a probabilistic model justify strategic action?
- What level of uncertainty is acceptable in national security?

3 GROUP STRUCTURE

1. **Chief Justice**
2. **Prosecutor**
3. **Defense Attorney**
4. **Dr. Ethan Cole (Lead Data Scientist – Defendant)**
5. **Independent Statistician (Expert Witness)**

1. Chief Justice

Your mission: Ensure fairness and deliver a reasoned verdict.

You must:

- Control speaking time
- Rule on objections
- Ask clarifying questions
- Deliver a structured final decision

You should evaluate:

- Technical reasoning
- Risk assessment
- Legal responsibility
- Ethical implications

In your final verdict, you must explain:

- Whether negligence occurred
- Whether the threshold choice was justified
- Whether responsibility lies with the individual or the institution

2. Prosecutor

Your position: The threshold was irresponsibly designed.

You must argue that:

- 87% was too low for autonomous escalation
- The cost of false positives was underestimated
- No sufficient human override mechanism existed
- The risk model ignored worst-case civilian consequences

You should question:

- How was the threshold mathematically justified?
- Were sensitivity and specificity properly balanced?
- Was Bayesian updating applied correctly?
- Were cascading risks evaluated?

Goal: Prove criminal negligence.

3. Defense Attorney

Your position: The system was scientifically and strategically justified.

You must argue that:

- In national security, false negatives are more dangerous than false positives
- Risk thresholds are policy decisions, not purely mathematical ones
- The model was statistically sound
- Perfect certainty is impossible

You should highlight:

- Decision theory principles
- Asymmetric cost functions
- Strategic deterrence logic
- Institutional responsibility

Goal: Create reasonable doubt.

4. Dr. Ethan Cole (Defendant – Lead Data Scientist)

You designed ARES.

You must explain:

- How the probabilistic model works
- Why 87% was selected
- What training data were used
- What safeguards were embedded
- Whether human override was technically feasible

Be precise.

Use technical vocabulary:

- confusion matrix
- Type I / Type II errors
- risk minimization
- algorithmic threshold calibration
- probabilistic confidence interval

You must defend your scientific integrity.

5. Independent Statistician (Expert Witness)

You are neutral.

You must explain clearly:

- Trade-offs between false positives and false negatives
- Threshold optimisation methods
- Risk modelling limitations
- Statistical uncertainty

You may:

- Support or weaken either side
- Criticize methodological assumptions
- Question robustness of the model

Your role is analytical, not emotional.

5 TRIAL STRUCTURE

Opening Statement – Prosecutor (3 min)

1. Opening Statement – Defense (3 min)
2. Direct Examination of Dr. Cole
3. Cross-Examination
4. Expert Witness Testimony
5. Closing Arguments
6. Judge's Verdict (reasoned, structured)

6 REQUIRED VOCABULARY

Mathematical

- probabilistic modelling
- threshold calibration
- sensitivity / specificity
- confidence interval
- decision theory
- cost asymmetry
- false positive / false negative

Legal

- criminal negligence
- liability
- reasonable doubt
- duty of care
- proportionality

Strategic

- national security
- escalation control
- deterrence
- systemic risk